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Experiment No-01

Experiment Name: Write and execute a machine-language program in MTS86 trainer board to perform the logical AND operation:

A AND B=? , A OR B= ?; Where A=02,B=A3.

Objective: The 8086 kit MTS86 is use to perform machine language program on it to solve any operations. This has I/O facilities and built in with different interfacing devices such as 8255, 8251, 8259, 8253, 8254 etc.

Specification:

- CPU: 8086
- Display Unit: LCD(16x2)
- RAM: 64kb
- ROM: 64kb
- User Memory: 64kb

Procedure: A AND B

Assembly Language	Machine Language
MOV AL, 02	B0, 02
AND AL, A3	24, A3

Truth Table:

A	B	A AND B
0	0	0
0	1	0
1	0	0
1	1	1

INPUT: A = 02

B = A3

OUTPUT: A = 02 → 0000 0010

B = A3 → 1010 0011

 AB = 02 → 0000 0010
Procedure: A OR B

Assembly Language	Machine Language
MOV AL, 02	B0, 02
OR AL, A3	OC, A3

Truth Table:

A	B	A OR B
0	0	0
0	1	1
1	0	1
1	1	1

INPUT: A = 02
 B = A3

OUTPUT: A = 02 → 0000 0010
 B = A3 → 1010 0011

 AB = 02 → 1010 0011

Experiment No-02

Experiment Name: Write and execute a machine language program in MTS86 trainer board to solve the logical NOT operation, $\hat{A} = ?$ Where A = 02.

Objective: The 8086 kit MTS86 is use to perform machine language program on it to solve any operations. This has I/O facilities and built in with different interfacing devices such as 8255, 8251, 8259, 8253, 8254 etc.

Specification:

- CPU: 8086
- Display Unit: LCD(16x2)
- RAM: 64kb
- ROM: 64kb
- User Memory: 64kb

Procedure:

Assembly Language	Machine Language
MOV AL, 02	B0, 02
NOT AL	F6, DO

Truth Table:

A	$\bar{A}$
0	1
1	0

INPUT:

A = 02

OUTPUT:A = 02 $\rightarrow$ 0000 0010

 $\bar{A}$ = FD $\rightarrow$ 1111 1101

Experiment No-03

Experiment Name: Write and execute a machine language program in MTS86 trainer board to perform the logical XOR operation $A \oplus B = ?$
Where A = 06 and B = 5E.

Objective: The 8086 kit MTS86 is use to perform machine language program on it to solve any operations. This has I/O facilities and built in with different interfacing devices such as 8255, 8251, 8259, 8253, 8254 etc.

Specification:

- CPU: 8086
- Display Unit: LCD(16x2)
- RAM: 64kb
- ROM: 64kb
- User Memory: 64kb

Procedure:

Assembly Language	Machine Language
MOV AL, 06H	B0, 06
MOV BL, 5EH	B3, 5E
XOR AL, BL	30, D8

Truth Table:

A	B	$A \oplus B$
0	0	0
0	1	1
1	0	1
1	1	0

INPUT: $A = 06 \rightarrow 0000\ 0110$ $B = 5E \rightarrow 0101\ 1110$ **OUTPUT:** $A = 06 \rightarrow 0000\ 0110$ $B = 5E \rightarrow 0101\ 1110$

 $AB = 02 \rightarrow 0101\ 1000$

Experiment No-06

Experiment Name: Write and execute a program in MTS86 trainer board and show hexadecimal value (0,1, ...F) on the seven-segment display.

Objective: The 8086 kit MTS86 is use to perform machine language program on it to solve any operations. This has I/O facilities and built in with different interfacing devices such as 8255, 8251, 8259, 8253, 8254 etc.

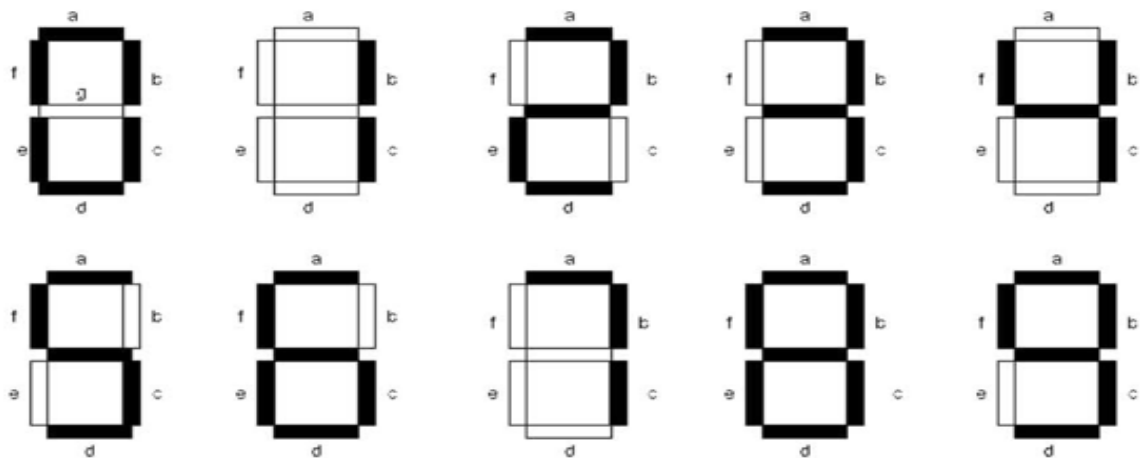
Specification:

- CPU: 8086
- Display Unit: LCD(16x2)
- RAM: 64kb
- ROM: 64kb
- User Memory: 64kb

Procedure:

Assembly Language	Machine Language
MOV AL,1000 0000B	B0, 80
OUT PORT-CR, AL	E6, 1F
MOV AL, 1111 0000B	B0, F0
OUT PORT-B, AL	E6, 1B
MOV AL, 0000 0000B	B0, 00
OUT PORT-C, AL	E6, 1D
MOV AL, C0	B0, C0
OUT PORT-A, AL	E6, 19

Code change for(0,1,2,9) as C0/F9/A4/B0/99/82/F8/80/F9.....

Example:

Experiment No-07

Experiment Name: Write and execute a program in MTS86 trainer board to display any value on the LED display.

Objective: The 8086 kit MTS86 is use to perform machine language program on it to solve any operations. This has I/O facilities and built in with different interfacing devices such as 8255, 8251, 8259, 8253, 8254 etc.

Specification:

- CPU: 8086
- Display Unit: LCD(16x2)
- RAM: 64kb
- ROM: 64kb
- User Memory: 64kb

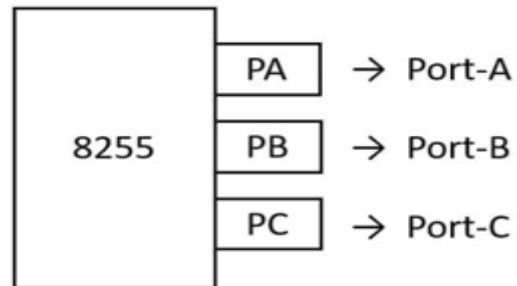
Procedure:

Assembly Language	Machine Language
MOV AL,1000 0000B	B0, 80
OUT PORT-CR, AL	E6, 1F
MOV AL, 1111 1111B	B0, FF
OUT PORT-B, AL	E6, 19
MOV AL, 0000 0000B	B0, 00
OUT PORT-C, AL	E6, 1D
MOV AL, 1111 0001B	B0, F1
OUT PORT-B, AL	E6, 1B

Code change (By SHL): 1111 0001B/1111 0010B/ 1111 0100B/ 1111 1000B

like this. By this way colour will change in different shape.

OUTPUT:



Experiment No-08

Experiment Name: Write and execute a program in MTS86 trainer board to interface and display output using the D/A converter.

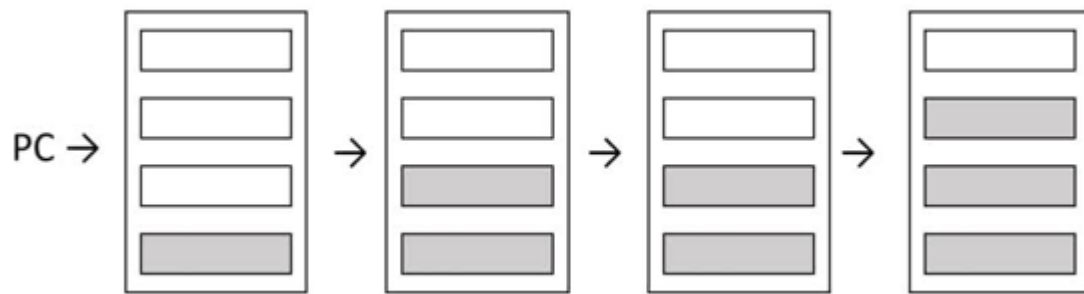
Objective: The 8086 kit MTS86 is use to perform machine language program on it to solve any operations. This has I/O facilities and built in with different interfacing devices such as 8255, 8251, 8259, 8253, 8254 etc.

Specification:

- CPU: 8086
- Display Unit: LCD(16x2)
- RAM: 64kb
- ROM: 64kb
- User Memory: 64kb

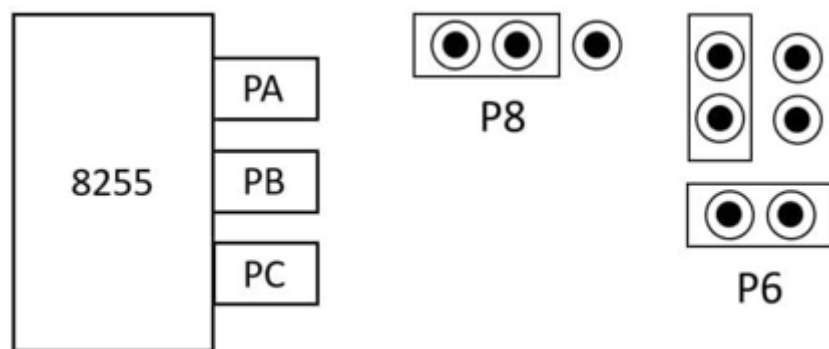
Procedure:

Assembly Language	Machine Language
MOV AL,1000 0000B	B0, 80
OUT PORT-CR, AL	E6, 1F
MOV AL, 1111 1111B	B0, FF
OUT PORT-B, AL	E6, 19
MOV AL, 1111 0000B	B0, F0
OUT PORT-B, AL	E6, 1B
MOV AL, 00 → 05	B0, 00
OUT PORT-C, AL	E6, 1D



Code change: 00/ 01/ 05/ 10/ 0F

By this coding the light level will increase or decrease.



Experiment No-09

Experiment Name: Write and execute a program in MTS86 trainer board to display output on the LED MATRIX.

Objective: The 8086 kit MTS86 is use to perform machine language program on it to solve any operations. This has I/O facilities and built in with different interfacing devices such as 8255, 8251, 8259, 8253, 8254 etc.

Specification:

- CPU: 8086
- Display Unit: LCD(16x2)
- RAM: 64kb
- ROM: 64kb
- User Memory: 64kb

Procedure:

Assembly Language	Machine Language
MOV AL,1000 0000B	B0, 80
OUT PORT-CR, AL	E6, 1F
MOV AL, 1111 1111B	B0, FF
OUT PORT-A, AL	E6, 19
MOV AL, 1111 0000B	B0, F0
OUT PORT-B, AL	E6, 1B
MOV AL, 0000 0001B	B0, 01
OUT PORT-C, AL	E6, 1C